Preliminary Evaluation of IMU-Based Gait Event Detection Algorithms for Rehabilitation-Oriented Gait Assessment

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Introduction

Wearable sensing technologies are increasingly used to support objective motor function assessment in gait rehabilitation, particularly in post-stroke and spinal cord injury (SCI) patients. Conventional instrumented gait analysis is accurate but relies on dedicated equipment, controlled environments and trained personnel, limiting frequent longitudinal monitoring. In this context, the NIMBLE-NEXT project aims to integrate multilevel longitudinal metrics for advanced robotic gait rehabilitation, including a wearable monitoring system for quantitative motor function assessment. Since reliable IMU-derived gait metrics depend on accurate heel strike (HS) and toe off (TO) detection, this study aimed to evaluate the agreement between four IMU-based gait event detection algorithms and an insole pressure sensor (IPS) reference using Bland-Altman analysis.

Methods

From the MovePort dataset [1], level treadmill walking trials at 2 km/h were selected. After excluding two healthy subjects with unreliable IPS-based event detection, 22 participants were retained. Synchronized IMU, IPS, EMG and motion capture recordings were available, with the IPS used as the reference system. HS and TO events were identified from total vertical force using a 50 N threshold, corresponding to upward and downward crossings; plausible cycles with stance duration $\geq$ 200 ms were retained. Four literature-based algorithms were implemented: Alg1, foot IMU using vertical acceleration and sagittal angular velocity [2]; Alg2, foot IMU using vertical acceleration [3]; Alg3, waist IMU using filtered trunk acceleration [4]; and Alg4, waist IMU using continuous wavelet transform-based processing [5]. IMU-derived events were matched to IPS events and HS and TO timing errors were computed separately. Subject-level outlier analysis was performed to identify extreme mismatches. Detected events were then used to compute temporal, spatiotemporal, phase/support, and symmetry metrics. Accuracy was further assessed through errors in the derived gait parameters; agreement with the reference was evaluated using Bland-Altman analysis on subject-level median values, derived from matched gait cycles.

Results

Temporal error boxplots showed that algorithms based on a foot IMU outperformed approaches based on a waist IMU, with Alg2 achieving the lowest median errors and variability for stride and stance time. Stance time exhibited greater variability than stride time, reflecting the higher sensitivity of TO detection errors. Bland–Altman analysis confirmed very good agreement between Alg2 and IPS for stride time, with minimal bias and most observations falling within the limits of agreement (Figure 1). S21 showed marked left-side HS over-detection.

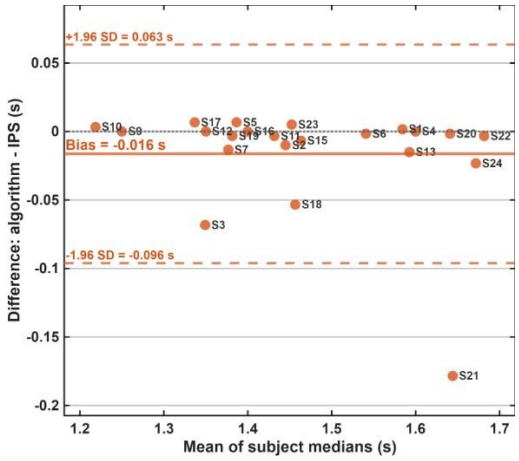


Figure 1: Bland–Altman plot comparing subject-level median stride time estimated by Alg2 with the IPS reference (n=22 participants).

Conclusions

Alg2 demonstrated the best overall preliminary performance for gait event detection and temporal parameter estimation. These findings support its preliminary selection for subsequent validation in post-stroke and SCI populations within the NIMBLE-NEXT wearable assessment pipeline. Future work will extend the analysis to the full set of gait metrics during rehabilitation-oriented and home-based assessments.

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